

1) Student Management System

START

PRIVATE name, rollno, marks

METHOD acceptDetails()

INPUT name

INPUT rollno

INPUT marks

END METHOD

METHOD calculateAverage()

average = marks

RETURN average

END METHOD

METHOD calculateAverage()

average = marks

RETURN average

END METHOD

METHOD displayResult()

IF marks >= 40 THEN

PRINT "PASS"

ELSE

PRINT "FAIL"

ENDIF

END METHOD

END CLASS

CREATE Student S

CALL S.acceptDetails()

CALL S.calculateAverage()

CALL S.displayResult()

STOP

2) Library Book Management.

START

CLASS Book

PRIVATE title, author, ISBN, isAvailable

METHOD issueBook()

IF isAvailable = TRUE THEN

isAvailable = FALSE

PRINT "Book Issued"

ELSE

PRINT "Book not available"

END IF

END METHOD

METHOD returnBook()

isAvailable = TRUE

PRINT "Book Returned"

END METHOD

METHOD displayDetails()

PRINT title

PRINT author

PRINT ISBN

PRINT isAvailable

END METHOD

END CLASS

CREATE book1

CREATE book2

CREATE book3

CALL issueBook()

CALL returnBook()

CALL displayDetails()

STOP

3) Employee Salary Calculation.

START

CLASS Employee

PRIVATE empID, name, basicPay, designation

METHOD calculateGross()

HRA = basicPay * 20/100

DA = basicPay * 10/100

gross = basicPay + HRA + DA

RETURN gross

END METHOD

METHOD calculateNet()

gross = calculateGross()

PF = gross * 12/100

TAX = gross * 5/100

net = gross - PF - TAX

RETURN net

END METHOD

METHOD displayPaySlip()

PRINT empID

PRINT name

PRINT designation

PRINT gross

PRINT net

END METHOD

END CLASS

CREATE Employee

CALL calculateGross()

CALL calculateNet()

CALL calculateNet()

CALL displayPaySlip()

STOP

4) shape area calculator using polymorphism

START

CLASS Shape

METHOD calculateArea()

END METHOD

END CLASS

CLASS Circle EXTENDS Shape

radius

METHOD calculateArea()

area = $3.14 * \text{radius} * \text{radius}$

PRINT area

END METHOD

END CLASS

CLASS Rectangle EXTENDS Shape

length, breadth

METHOD calculateArea()

area = length * breadth

PRINT area

END METHOD

END CLASS

CLASS Triangle EXTENDS Shape

base, height

METHOD calculateArea()

area = $0.5 * \text{base} * \text{height}$

PRINT area

END METHOD

END CLASS

CREATE Circle

CREATE Rectangle

CREATE Triangle

CALL calculateArea()

STOP()

5) Vehicle Registration System

START

CLASS Vehicle

RegNo

owner

FuelType

METHOD displayDetails()

PRINT RegNo

PRINT owner

PRINT FuelType

END METHOD

END CLASS

CLASS TwoWheeler EXTENDS Vehicle

engineCapacity

METHOD displayDetails()

PRINT RegNo

PRINT owner

PRINT FuelType

PRINT engineCapacity

END METHOD

END CLASS

CLASS FourWheeler EXTENDS Vehicle

seatingCapacity

ACStatus

METHOD displayDetails()

PRINT RegNo

PRINT owner

PRINT FuelType

PRINT seatingCapacity

PRINT ACStatus

END METHOD

END CLASS

CREATE TwoWheeler()

CREATE FourWheeler()

CALL displayDetails()

STOP

6) Bank Account Hierarchy.

START

CLASS BankAccount

 accno

 holder

 balance

 METHOD withdraw (amount)

 END METHOD

END CLASS

CLASS SavingsAccount EXTENDS BankAccount

 METHOD withdraw (amount)

 IF balance $\geq$ amount THEN

 balance = balance - amount

 PRINT "Withdrawal Successful"

 ELSE

 PRINT "Insufficient Balance"

 END IF

 END METHOD

END CLASS

CLASS CurrentAccount EXTENDS BankAccount

 overdraftlimit

 METHOD withdraw (amount)

 IF balance + overdraftlimit $\geq$ amount THEN

 balance = balance - amount

 PRINT "Withdrawal Successful"

 ELSE

 PRINT "Limit Exceeded"

 END IF

 END METHOD

END CLASS

CREATE SavingsAccount

CREATE CurrentAccount

CALL withdraw ()

STOP

7) Hospital Patient Record System

START

CLASS Patient

PRIVATE name

PRIVATE diagnosis

METHOD generateBill ()

END METHOD

END CLASS

CLASS InPatient EXTENDS Patient

ward no

daily charge

days

METHOD generateBill ()

$bill = \text{daily charge} \times \text{days}$

PRINT bill

END METHOD

END CLASS

CLASS OutPatient EXTENDS Patient

consultation Fee

METHOD generateBill ()

PRINT consultation Fee

END METHOD

END CLASS

CREATE InPatient

CREATE OutPatient

CALL generateBill ()

STOP

8) E-Commerce Product Catalog

START

CLASS Product

name

price

METHOD applyDiscount ()

END METHOD

END CLASS

CLASS Electronics EXTENDS Product

METHOD applyDiscount ()

price = price - (price * 10/100)

END METHOD

END CLASS

CLASS Clothing EXTENDS Product

METHOD applyDiscount ()

price = price - (price * 20/100)

END METHOD

END CLASS

CLASS Grocery EXTENDS Product

METHOD applyDiscount ()

price = price - (price * 5/100)

END METHOD

END CLASS

CREATE Electronics

CREATE Clothing

CREATE Grocery

CALL applyDiscount ()

STOP

9) University Staff Hierarchy

START

CLASS Staff

PRIVATE StaffId

PROTECTED Salary

METHOD Calculate Allowance ()

END METHOD

END CLASS

CLASS TeachingStaff EXTENDS Staff

METHOD Calculate Allowance ()

allowance = Salary * 20/100

END METHOD

END CLASS

CLASS Professor EXTENDS Staff

METHOD Calculate Allowance ()

allowance = Salary * 30/100

END METHOD

END CLASS

CLASS Lecturer EXTENDS TeachingStaff

METHOD Calculate Allowance ()

allowance = Salary * 10/100

END METHOD

END CLASS

CLASS NonTeachingStaff EXTENDS Staff

METHOD Calculate Allowance ()

allowance = Salary * 10/100

END METHOD

END CLASS

CREATE Professor

CREATE Lecturer

CREATE NonTeachingStaff

CALL Calculate Allowance ()

STOP

10) Smart Home Device Controller

START

CLASS Smart Device

METHOD execute Command (Command)

END METHOD

END CLASS

CLASS SmartLight EXTENDS Smart Device

METHOD execute Command (Command)

IF Command = "on" THEN

PRINT = "Light on"

ELSE IF Command = "off" THEN

PRINT = "Light off"

ELSE

PRINT "Light Status"

END IF

END METHOD

END CLASS

CLASS SmartThermostat EXTENDS Smart Device

METHOD execute Command (Command)

PRINT "Thermostat Executed"

END METHOD

END CLASS

CLASS SmartLock EXTENDS Smart Device

PRIVATE PIN

METHOD execute Command (Command)

PRINT "Lock Executed Successfully"

END METHOD

END CLASS

CREATE SmartLight

CREATE SmartThermostat

CREATE SmartLock

CALL Execute Command ()

STOP